

```

1  package uart_tb_pkg;
2
3  class uart_txn;
4      rand bit [7:0] data;
5      bit [9:0] frame; // start + data + stop
6
7      function void build_frame();
8          frame = {1'b1,data,1'b0}; // stop, data, start
9      endfunction
10
11      function void display(string tag);
12          $display("[%s] Data=%0h Frame=%b",tag, data, frame);
13      endfunction
14  endclass
15
16  class generator;
17      mailbox #(uart_txn) gen2drv;
18
19      function new(mailbox #(uart_txn) mb);
20          gen2drv = mb;
21      endfunction
22
23      task run(int num_txns=5);
24          uart_txn t;
25          repeat (num_txns) begin
26              t = new();
27              assert(t.randomize());
28              t.build_frame();
29              t.display("GEN");
30              gen2drv.put(t);
31          end
32      endtask
33  endclass
34
35  class driver;
36      virtual uart_if vif;
37      mailbox #(uart_txn) gen2drv;
38
39      function new(virtual uart_if vif, mailbox #(uart_txn) mb);
40          this.vif = vif;
41          this.gen2drv = mb;
42      endfunction
43
44      task run(int baud_rate=16);
45          uart_txn t;
46          forever begin
47              gen2drv.get(t);
48              t.display("DRV");
49              foreach(t.frame[i]) begin
50                  vif.rx <= t.frame[i];
51                  repeat(baud_rate) @(posedge vif.clk);
52              end
53              vif.rx <= 1; //idle
54          end
55      endtask
56  endclass
57
58  class monitor;
59      virtual uart_if vif;
60      mailbox #(uart_txn) mon2sb;
61
62      function new(virtual uart_if vif, mailbox #(uart_txn) mb);
63          this.vif = vif;
64          this.mon2sb = mb;
65      endfunction
66
67      task run();
68          uart_txn t;
69          forever begin

```

```

70         @(posedge vif.data_valid);
71         t = new();
72         t.data = vif.dout;
73         t.build_frame();
74         t.display("MON");
75         mon2sb.put(t);
76     end
77 endtask
78 endclass
79
80 class scoreboard;
81     mailbox #(uart_txn) mon2sb;
82     mailbox #(uart_txn) gen2sb;
83
84     function new(mailbox #(uart_txn) m1, mailbox #(uart_txn) m2);
85         this.gen2sb = m1;
86         this.mon2sb = m2;
87     endfunction
88
89     task run();
90         uart_txn t_gen, t_mon;
91         forever begin
92             gen2sb.get(t_gen);
93             mon2sb.get(t_mon);
94             if (t_gen.data != t_mon.data)
95                 $error("[SB] MISMATCH Sent=%0h Recived=%0h",t_gen.data,t_mon.data);
96             else
97                 $display("[SB] PASS Data=%0h",t_gen.data);
98         end
99     endtask
100 endclass
101
102 endpackage

```